

Two Paths Out of Imbalance: How Competing Exits Produce Persistence in Interstate Relations, 1816–2012

Journal Title
XX(X):1–20
©The Author(s) 2026
Reprints and permission:
sagepub.co.uk/journalsPermissions.nav
DOI: 10.1177/ToBeAssigned
www.sagepub.com/

SAGE

[Author information removed for peer review]

Abstract

Why do some inconsistent configurations of interstate alliances and hostilities persist while others rapidly dissolve or realign? Structural balance theory predicts pressure toward relational consistency, but it does not specify which of two routes out of imbalance — a relationship dissolving, or being retained and realigned into balance — a given episode will take, or how long that will take. Using signed interstate networks built from Correlates of War alliance and militarized-dispute data, 1816–2012, I identify 6,277 spells of triadic imbalance and model which route each takes with competing-risks hazard models. Greater triadic embeddedness sharply reduces the hazard of dissolution while increasing the hazard of realignment, so persistence concentrates at intermediate embeddedness, where neither pathway dominates. Within realigning triads, the tie carrying the greatest unbalanced load is disproportionately the one that changes, beyond its current sign and its own embeddedness alone. Persistence is thus an emergent consequence of competing resolution routes rather than an absence of balance-restoring pressure: interstate hostilities and alliance commitments are shaped by the triadic structures in which bilateral relations are embedded.

Keywords

structural balance, signed networks, militarized disputes, alliance networks, competing risks

1. Introduction

International alliances and militarized disputes do not arrange themselves randomly. A state's formal defense commitments and its militarized disputes are relationally structured: which other states an actor is allied with and which it is locked in confrontation with are rarely independent of one another. When a state finds itself simultaneously allied with two states that are themselves rivals, international relations scholars have long treated the resulting configuration as structurally unstable — a source of relational pressure that should, sooner or later, resolve (Maoz et al. 2007). That intuition has a formal foundation in structural balance theory, a theory of relational change: inconsistent configurations generate pressures that should alter the relationships composing them and move social structure towards greater balance (Heider 1946; Cartwright and Harary 1956). Yet the empirical evidence commonly offered for balance in interstate networks, as elsewhere, is much less dynamic than the theory it supports. Studies frequently establish that balanced configurations are more prevalent than unbalanced ones, or that observed networks contain more balance than appropriate null models would

predict (Davis 1967; Maoz et al. 2007). Even this narrower, purely inferential question has grown more demanding on its own terms: whether a signed network appears consistent with balance theory at all can depend on which null model is used and whether it respects heterogeneity across actors (Gallo et al. 2024). But however that inferential question is resolved, prevalence evidence of this kind, however carefully measured, does not show what happens to a particular configuration once it becomes unbalanced. Longitudinal evidence tracing these transitions is considerably scarcer and more mixed: where it exists, it has generally found that unbalanced configurations do not move smoothly toward balance, but can persist, oscillate, or resolve through mechanisms the classical model does not specify (Doreian and Krackhardt 2001). A basic dynamic question at the heart of balance theory therefore remains surprisingly open: what path does imbalance take once it emerges?

Not every unbalanced configuration immediately becomes balanced. Some disappear quickly, others remain inconsistent for years, and still others undergo substantial relational realignment. The persistence of imbalance is particularly important. If inconsistency generates pressure for change,

why are some actors apparently able to sustain relational configurations that remain inconsistent over extended periods? Persistence need not imply the absence of balance pressures. It may instead reflect differences in the relational opportunities available for acting on them. Understanding persistent imbalance therefore requires following unbalanced configurations through time rather than inferring their dynamics from the relative prevalence of balanced and unbalanced states.

Doing so reveals a second issue. There is more than one way for imbalance to end, and two contrasting configurations drawn directly from the data analyzed below illustrate it. In 1940, Britain, Germany, and the Soviet Union formed an unbalanced triad in which all three pairwise relationships were mutually hostile. Britain and Germany were already at war. Britain's own militarized dispute with the Soviet Union dated to September 1939, when the Soviet invasion of eastern Poland — coordinated with Nazi Germany under the Molotov-Ribbentrop Pact's secret protocol — was followed by a second dispute in January 1940, as Britain and France weighed intervention against the Soviet invasion of Finland. Germany and the Soviet Union, meanwhile, had their own active militarized dispute despite the Pact's formal nonaggression terms — an all-hostile configuration unbalanced by the same criterion as a lone dissenting tie. This lasted exactly one year. In June 1941, Germany's invasion of the Soviet Union escalated the Germany-USSR dispute to full war, and within weeks Britain and the Soviet Union set aside their own dispute and signed a mutual assistance agreement — the tie that changed was specifically Britain-USSR, flipping from hostile to allied, while Britain-Germany remained at war and Germany-USSR remained hostile. The triad resolved through realignment: one relationship changed sign while the other two persisted. Contrast this with the United States, France, and the Soviet Union during the late Cold War. Throughout 1977–1986, the US-France NATO alliance and a France-USSR consultative entente were both positive while the US and USSR remained locked in a continuous decade-long series of militarized disputes: an inconsistent configuration that, unlike the Britain-Germany-USSR case, did not resolve quickly. It persisted for ten years, until 1987, when it disappeared not because the two superpowers reconciled but because that specific dispute series simply stopped being renewed amid the broader U.S.-Soviet thaw. This was resolution through dissolution: a relationship disappeared rather than changed. Why does imbalance sometimes resolve within a year through realignment and sometimes take a decade through dissolution instead?

This distinction changes the question we should ask of balance dynamics. The central problem is not simply whether an unbalanced configuration eventually becomes balanced. It is to explain its trajectory: how long imbalance persists and, when it ends, through which relational process it is resolved. Imbalance can terminate through very different paths, and not all of them lead to balance. A configuration may cease to exist because one of its relationships disappears, exiting the domain in which balance is even defined, or it may remain intact while a relationship is transformed, resolving into balance. Looking only at whether balance ultimately obtains obscures this difference. It also obscures the possibility that persistent imbalance is generated not by a distinct force that preserves inconsistency, but by the relative availability of the alternative routes through which inconsistency can end.

I argue that these trajectories are structured by relational embeddedness: how densely a relationship is woven into the surrounding network. Embeddedness should have opposing effects on the two routes out of imbalance. A weakly embedded relationship is not reinforced by much else, so its disappearance disrupts little beyond the focal relationship, leaving dissolution a readily available exit. A densely embedded relationship is costlier to lose for exactly the reason it is harder to abandon: removing it would disturb everything else built on the same foundation, but that same density means it also sits inside more relational configurations whose consistency depends on its sign, making realignment comparatively more available once the tie is retained. As embeddedness rises, then, dissolution should become less likely while realignment becomes more likely; Section 2 develops why each half of this claim holds.

This produces a less obvious implication for persistence. If weakly embedded configurations have an accessible route out through dissolution, they need not remain unbalanced for long. If highly embedded configurations are unlikely to dissolve but increasingly likely to realign, they too should tend to leave the unbalanced state. The configurations most likely to persist are therefore potentially those between these two resolution regimes: embedded enough that dissolution is no longer easy, but not so embedded that realignment reliably resolves the inconsistency. Persistence, on this account, is not simply the opposite of structural pressure towards balance. It emerges from the intersection of two distinct processes for leaving imbalance.

This argument calls for a different empirical approach to structural balance. Rather than comparing the prevalence of balanced and unbalanced configurations, I identify spells of imbalance and follow their trajectories. I construct yearly

signed networks of the interstate system from Correlates of War alliance and militarised-dispute data between 1816 and 2012. Across 378,623 closed triad-years, I identify 6,277 distinct spells in which a triad enters and remains in an unbalanced state. A spell continues for as long as the same triad remains closed and unbalanced and ends when a constituent relationship disappears, when the triad realigns into balance, or when observation ends.

This structure makes persistence and resolution parts of the same dynamic process. In every year that an unbalanced triad survives, it remains at risk of two competing transitions: dissolution or realignment. I therefore estimate discrete-time cause-specific hazard models with time-varying measures of network structure. The models ask, for each year an unbalanced configuration remains at risk, how its embeddedness changes the probability that it dissolves in the following year and, separately, the probability that it realigns. Continued imbalance is not treated as a third kind of resolution. It is duration in the origin state: another year in which neither competing transition has yet occurred.

The results reveal a striking structural separation between the two pathways. Greater triadic embeddedness sharply reduces the hazard of dissolution while increasing the hazard of realignment. A one-standard-deviation increase in embeddedness is associated with roughly a 92 per cent reduction in the odds of dissolution but a 55 per cent increase in the odds of realignment in the following year — an effect substantially, though not entirely, driven by formal multilateral institutionalization, since the great majority of the network's most embedded ties sit inside a small number of major 20th- and 21st-century alliance blocs (Section 4.5). These opposing hazards produce distinct trajectories of imbalance. Weakly embedded configurations predominantly disappear through dissolution. Highly embedded configurations overwhelmingly resolve through realignment. Intermediate configurations exhibit the greatest persistence because the dissolution pathway has already narrowed while realignment has not yet become as dominant.

The analysis also shows that this logic extends within realigning triads: the tie carrying the greatest unbalanced load is disproportionately the one that changes, beyond what its sign alone would predict.

The paper makes a shift in emphasis rather than proposing an alternative to structural balance theory: understanding the dynamics implied by that theory requires looking inside the transition out of imbalance. Network structure shapes not merely whether imbalance ends, but the pathway and pace through which it does so, a claim developed in Section 2

against balance theory's own dynamic formulations, a wider IR literature on extra-dyadic network interdependence, and the dyadic rivalry- and alliance-termination literatures.

2. Theory: Dissolution and Realignment as Competing Routes Out of Imbalance

2.1 From balance outcomes to pathways of resolution

Structural balance theory holds that triads of signed relationships are stable when balanced — all positive, or two negative and one positive — and unstable when not, generating pressure toward consistency (Heider 1946; Cartwright and Harary 1956), part of a broader contemporary signed-networks field concerned with how negative and positive ties jointly structure social and political life, international relations included (Harrigan, Labianca, and Agneessens 2020; Estrada 2019). Dynamic extensions model this pressure as a stochastic process in which unbalanced triads flip a tie's sign until the network reaches balance (Hummon and Doreian 2003; Antal, Krapivsky, and Redner 2005; Marvel et al. 2011), and applications to interstate relations have used exactly this logic to explain alliance formation and the polarization of rivalry blocs (Maoz et al. 2007). These models identify one distinct reason imbalance can persist: a network can become “jammed” in a configuration from which no single available sign change reduces total imbalance, so persistence reflects a local optimum of the balance-seeking dynamics themselves (van de Rijt 2011). The persistence mechanism developed below is different, and complementary, to this: it does not depend on any such local-optimum property, but on the joint unavailability, at a given moment, of the two concrete exits — dissolution and realignment — that the surrounding network structure makes available or forecloses.

These prominent dynamic models operationalize adjustment specifically as sign-switching: a tie is treated as a fixed feature of the relationship space, present and available to be flipped, with modeled transitions running only in that direction. This is a modeling choice within particular dynamic formulations, not an inherent limit of balance theory or the broader signed-networks tradition, which can and does accommodate null or absent ties (e.g., Doreian and Krackhardt 2001) and, in the interstate case, has been extended to separate the probability that two states interact at all from the conditional sign of that interaction (Lerner 2016), to a dynamic exponential-random-graph framework that treats interstate relations as positive, negative, or absent under an

explicit balance-theoretic specification (Fritz et al. 2025), and, most recently, to separable models distinguishing the interaction process from the sign process altogether (Caimo and Gollini 2026). What remains underspecified is not whether ties can disappear but *when* a concrete episode of triadic imbalance resolves through a tie disappearing rather than through realignment. Alliances lapse, and militarized disputes are not indefinitely ongoing threats — both are coded as *active* commitments in a given year, and the absence of renewal is itself informative — but no existing account predicts which of the two resolutions a given unbalanced triad will take, or when.

The closest prior attempts to track this directly in interstate data establish how much balance a signed international system exhibits and how that quantity moves over time, rather than following individual triads as spells with a start, a duration, and a specific terminal event. Doreian and Mrvar's (2015) study of the signed international system from 1946 to 1999 shows that system-wide imbalance does not decline monotonically; Oishi and Sakuwa (2025), tracking a longer alliance-and-rivalry network from 1816 to 2009, find that consistency with balance theory itself varies substantially across historical periods; and Diaz-Diaz, Bartesaghi, and Estrada (2024) develop a local balance index and use its fluctuations from 1816 to 2014 to identify historically salient episodes. None of these studies asks, for a given closed triad that has become unbalanced, how long that specific episode persists or which of the two available exits ends it.

A broader IR network literature has already established that alliance and conflict behavior is not well described as a set of independent dyadic choices, a point recent syntheses treat as largely settled across conflict and cooperation alike (Larson 2021). Warren's (2010) evolving-network model of alliance formation shows that triadic patterns of amity and enmity, not dyadic considerations alone, shape which states ally with which; Cranmer, Desmarais, and Menninga (2012) model the complex, non-dyadic dependencies of the alliance network directly rather than treating dyads as conditionally independent, and Cranmer, Desmarais, and Kirkland (2012) show that closed alliance triangles generate their own synergy in the probability a new tie forms, beyond what the three dyads considered separately would predict; and Corbetta and Grant (2012) use structural balance itself to show that triadic configuration, not dyadic calculation, governs how third parties intervene in militarized disputes. Kinne's (2024) work on the network context of defense cooperation agreements extends the same tradition to a different institutional form, showing that an agreement's effectiveness depends on the surrounding network of related

commitments among third parties. This literature has not, however, asked what happens once a specific triad becomes internally inconsistent: it establishes that formation and intervention are extra-dyadically interdependent, not how that surrounding structure allocates a given episode of imbalance between the two available exits.

Two further, dyadic-level literatures study disappearance itself, on either side of the sign divide, without connecting it to that surrounding structure. On the negative side, research on rivalry termination documents that enduring interstate rivalries do eventually end, focusing on the conditions — power shifts, democratization, external threats, and the credibility that liberal domestic and international institutions confer — associated with termination (Bennett 1996; Diehl and Goertz 2000; Colaresi, Rasler, and Thompson 2007; Prins and Daxecker 2008; Sakuwa and Thompson 2019), and more recent work has tried to integrate the resulting proliferation of partial mechanisms into a more general account of how a given rivalry terminates when it does (Thompson 2022), with domestic institutional reform in either rival continuing to predict termination in the most recent data available (Saunders 2023). On the positive side, a parallel literature on alliance termination asks why states abrogate formal commitments, treating an alliance's end as a strategic choice conditioned on its terms and the parties' changing interests (Leeds and Savun 2007); more recent work qualifies even this dyadic framing somewhat by showing that an alliance's reliability is itself a function of the wider threat environment the two parties share, not the bilateral commitment considered in isolation (Nieman and Gibler 2023). Both remain, at core, dyadic theories of exit.

Combining these threads gives the paper its actual claim. IR network research has established that alliance and conflict behavior is extra-dyadically interdependent; rivalry- and alliance-termination research has established that both kinds of relationship do eventually end. What neither specifies is how the surrounding network structure allocates a given episode of triadic inconsistency between its two available exits, dissolution and realignment, or how long that allocation takes to resolve. Balance theory predicts *when* inconsistency arises; the exit literatures describe the two channels through which it can end without asking what triadic structure the relationship sits in; embeddedness, developed next, is the structural predictor that links the three.

2.2 Embeddedness reallocates resolution between dissolution and realignment

Consider an unbalanced triad — a state simultaneously allied with two others who are themselves rivals, say. Two structural properties of the ties involved determine what happens to this inconsistency.

In brief, embeddedness raises the cost of dissolution, since removing a tie disrupts every other relationship built on it, while, conditional on the tie surviving, it also multiplies the surrounding contexts in which the tie's sign matters, making realignment comparatively more available; the two effects pull dissolution and realignment hazards in opposite directions. This rests on how densely each tie is embedded in the surrounding network — a structural operationalization of Granovetter's (1985) argument that a relationship's own dynamics cannot be understood apart from the wider network surrounding it: how many other pairs of states are jointly connected to both endpoints of the tie, forming overlapping triangles. Granovetter supplies the broad intuition; a more specific mechanism comes from the literature on network closure and relational embeddedness that followed him. A tie embedded in a closed triad, where both parties also share a mutual third party, becomes what Krackhardt (1998) calls a Simmelian tie: qualitatively more constraining, and more difficult to sever, than an otherwise-equivalent tie with no such closure; outside interstate relations, this same structural embeddedness — as distinct from the mere existence or centrality of a relationship — has been shown directly to reduce the odds a relationship dissolves, in a study of interfirm joint ventures (Polidoro, Ahuja, and Mitchell 2011). Closure among a set of actors generates the kind of social capital that raises the cost of exit and lowers the cost of monitoring (Coleman 1988), and embeddedness carries a further, more particular cost: the same redundant ties that support trust and reduce transaction costs also mean that severing any one of them forfeits value built into the surrounding structure, not just into the focal relationship itself (Uzzi 1997). A weakly embedded tie is not reinforced by much else, so the path of least resistance when tension arises is simply for it not to persist. A densely embedded tie is different in what it costs to lose: removing it would disturb many other relationships built on the same foundation, since each overlapping triangle depends on the tie remaining in place regardless of whether that triangle is itself currently balanced or unbalanced. High embeddedness should therefore make dissolution a more consequential, and so less likely, choice of exit.

Conditional on the relationship being retained, a more embedded tie instead sits inside more surrounding configurations whose own consistency depends on its sign. This is a claim about exposure, not about instantaneous pressure: a tie's total embeddedness does not by itself measure how much consistency pressure currently bears on it, since a triangle that happens to be balanced generates no pressure at all regardless of how many such triangles surround the tie (Section 2.3 makes this precise). What total embeddedness does capture is how many surrounding configurations *could* turn unbalanced, and so generate pressure, as the rest of the network continues to change around a tie that is itself held fixed for now. A highly embedded tie that faces no pressure from any of its triangles today is correspondingly more exposed, over the course of the spell, to eventually facing pressure from one of them than a weakly embedded tie is, simply because it has more surrounding triangles whose own balance is not fixed. Conceptually, then, embeddedness and unbalanced load distinguish structural exposure from realized consistency pressure; Section 2.3 sharpens the latter into a tie-level prediction by isolating specifically the subset of a tie's surrounding triangles that are unbalanced at a given moment, rather than the full set embeddedness counts. Whether the two are empirically *separable*, rather than merely conceptually distinct, is a different question, taken up directly in Section 4.1: the interstate network turns out not to contain enough between-triad variation to identify them apart from one another, since the two measures are almost collinear across triads: their distinction becomes identifiable only *within* a triad, where edges share similar overall embeddedness but differ substantially in current unbalanced load. This prediction is opposite to a naive reading of “structural constraint prevents change”: embeddedness reduces the hazard of dissolution and increases the hazard of realignment; it does not suppress change generally.

The second property is the structural position of the actors involved, independent of any specific tie. Following Burt's (1992) concept of network constraint, an actor whose relationships are concentrated among a small, mutually interconnected set of partners has less independent latitude than one whose relationships span structurally disconnected parts of the network. Conceptually this differs from triadic embeddedness (an actor's overall position, not the redundancy of one relationship), but the two are not statistically independent: states central enough to broker many relationships, and therefore embedded in the most overlapping triangles, tend mechanically to register *low* Burt constraint, since their many partners are not all connected

to one another. This correlation ($r=-0.78$ below) means the two measures can produce opposite-looking estimates if either is included without the other, not because the theory is wrong about one, but because omitting one leaves the model unable to distinguish “this actor is unusually central” from “this actor’s ties happen to be concentrated.” Properly separated, the prediction for actor constraint parallels tie embeddedness: more constrained actors, like more embedded ties, should be less likely to see their relationships lapse and more likely to see them realigned.

2.3 Within a realigning triad, which tie changes?

The argument so far concerns which *triads* realign; a further question is which *tie* changes when one does. Overlapping structure matters here too, but not in its total, undifferentiated form: total structural embeddedness counts *every* overlapping triangle a tie belongs to, whether or not those triangles are themselves internally consistent, and a tie embedded in many already-balanced triangles is under no particular pressure from them, since a balanced triangle generates no consistency demand on its ties in the first place. Pressure to change should instead come specifically from the *other unbalanced* triads a tie is embedded in: each is another unbalanced configuration whose inconsistency would be resolved if this particular tie flipped, and a tie sitting at the intersection of many such triangles is pulled toward realignment from every one of them at once. This motivates a measure of *unbalanced load* — for a given tie, the number of other unbalanced triads (besides the focal one) that also contain it — and the corresponding prediction: within a realigning triad, the tie that changes should be the one with the highest unbalanced load, not the lowest total embeddedness or the highest betweenness.

This prediction has a close antecedent outside interstate relations. In a longitudinal study of interpersonal sentiment networks, Rawlings and Friedkin (2017) show that a directed sentiment’s own probability of conversion rises with its exposure to balance-rule violations across the triads containing it — in effect, the sentiment-network analogue of unbalanced load. Unbalanced load is accordingly not offered here as a new general balance-theoretic mechanism; the contribution is narrower and, for that reason, more demanding of the data. Rawlings and Friedkin ask whether greater exposure makes conversion more likely in the abstract; the present question is conditional on an unbalanced triad already resolving through realignment, which of its three candidate relationships is the one that changes, and whether the relative load carried by the alternatives predicts

that choice. The prediction is best read as an out-of-domain extension of an existing micro-mechanism, from small, densely observed interpersonal groups to a sparse, substantively costly interstate network of alliances and disputes, and from an unconditional conversion probability to a within-event choice among the three relationships actually available, rather than as a mechanism invented from scratch.

This is stated as a general prediction about which tie changes, independent of direction. It need not apply uniformly, however: an alliance-to-hostility realignment and a hostility-to-alliance realignment are substantively different kinds of relational change, and Leeds and Savun (2007) show alliance termination specifically is shaped by dyadic-level considerations a purely structural account need not capture. I therefore test the unbalanced-load prediction separately for the two directions rather than assuming the same mechanism governs both equally.

2.4 Hypotheses

- H1. Triadic embeddedness reduces the hazard of dissolution and increases the hazard of realignment to balance.
- H2. Where a triad does realign, the tie that changes is the one with the highest unbalanced load — the tie embedded in the most other unbalanced triads at that moment, and therefore under the greatest consistency pressure.

These two hypotheses give the paper its core structure: H1 asks which triads realign rather than dissolve; H2 asks which tie realigns when one does. A third, supplementary expectation follows at the level of the actor rather than the tie or triad: if embeddedness makes exit more consequential for a relationship, an analogous position-level property, how constrained an actor’s own relationships are (Burt 1992), should shape the same choice for the actor as a whole, reducing the hazard of dissolution and increasing the hazard of realignment once estimated jointly with triadic embeddedness. This is reported in Section 4.2 alongside a specification check rather than as a third numbered hypothesis, since it is not conceptually independent of H1 so much as a parallel manifestation at a different level of analysis, and a model omitting triadic embeddedness can recover the opposite sign on actor constraint entirely, reflecting the two measures’ substantial negative correlation rather than a contradiction in the theory.

3. Data and Research Design

3.1 Constructing the signed network and identifying spells

The empirical analysis draws on the Correlates of War (COW) project, the longest-running and most widely used data infrastructure in quantitative international relations (Singer and Small 1972; Sarkees and Wayman 2010), covering the interstate system from the Congress of Vienna to the present. I combine two COW datasets to build a signed dyad-year network, 1816–2012 (the alliance data’s own right-censoring point, adopted rather than the militarized-dispute data’s longer 1816–2014 coverage, since their union produces a data-coverage artifact in the final two years, where negative ties exist with essentially no positive ties to compare them against).

Positive ties. A dyad-year is coded positive if the two states have an active, formally ratified alliance commitment under the Formal Interstate Alliance dataset, v4.1 (Gibler 2009), which distinguishes four types ranked from strongest to weakest: *defense* pacts (obligating military support if a partner is attacked), *neutrality* pacts (committing a state not to join any adversary of its partner), *nonaggression* pacts (pledging signatories not to attack one another), and *ententes* (the loosest category, committing signatories to consult one another in a crisis). All four are coded as a positive tie here, following standard practice in the balance-theoretic literature (e.g., Maoz et al. 2007): what matters for the balance calculation is that the relationship is a formally recognized, publicly committed alignment, not the specific military obligation it carries.

Negative ties. A dyad-year is coded negative if the two states are parties to a militarized interstate dispute (MID) at hostility level 3 or higher, from the Dyadic MIDs dataset, v4.03 (Maoz et al. 2019). A MID is an explicit, government-authorized threat, display, or use of military force between COW system members, short of full-scale war; COW codes each dispute’s peak hostility on a five-point ordinal scale (1 = no militarized action, 2 = threat to use force, 3 = display of force, 4 = use of force, 5 = war). Restricting to level 3 and above excludes disputes that remained at the level of verbal threat and retains only cases involving an actual militarized display, use of force, or war — the threshold most commonly used in the conflict literature to distinguish a genuine militarized confrontation from diplomatic friction.

A dyad with neither an active alliance nor a qualifying dispute in a given year has no signed tie and does not contribute to that year’s network; no dyad-year in the observed data has both simultaneously.

For each year, I enumerate all closed triads — triples of states for which all three pairwise ties are signed — using triangle enumeration on the year’s signed graph, following standard practice in the signed-networks literature for restricting analysis to fully-determined configurations (e.g., Leskovec, Huttenlocher, and Kleinberg 2010). This yields 378,623 triad-years spanning 175 states, of which 97.3% are balanced in the Cartwright-Harary sense (product of the three edge signs positive) and 2.73% unbalanced. The primary analysis adopts this strong-balance criterion, under which both the two-positive/one-negative and the all-negative configuration are unbalanced; Davis’s (1967) weak-balance criterion instead treats the all-negative configuration as balanced, since three mutual enemies need not all belong to the same faction, and recent signed-network evidence shows this distinction can matter empirically (Gallo et al. 2024; Fritz et al. 2025). All-negative triads are a small share of the unbalanced population analyzed here (297 of 10,350 unbalanced triad-years, 2.87%), and the online appendix (Section A14) reclassifies them as balanced and re-estimates every hazard model in Table 2 on the resulting spell structure: the embeddedness effect on both hazards is materially unchanged in sign, magnitude, and significance. Large multilateral alliances — coded, like any alliance, as a complete dyadic clique among their signatories — contribute a disproportionate share of the closed-triad population throughout the panel, from the 10-member German Confederation before 1866 to NATO, the Warsaw Pact, and the Rio Pact after 1945; this is a real feature of how multilateral treaties enter the data, not a coding error, and the online appendix (Sections A7 and A2) shows the paper’s central results are robust to it, though not entirely independent of it.

Tracking each triad.id across consecutive years identifies spells of triadic imbalance: a spell begins when a triad becomes unbalanced (having been balanced, or newly formed as a closed triad) and continues through consecutive years in which it remains a closed, unbalanced triad. A spell ends in one of three ways, observable directly from the data: the triad disappears from the closed-triad population entirely (dissolution — one of its three ties lapsed), the triad reappears as balanced (realignment), or the observation window ends while the triad is still unbalanced and intact (right-censoring, 180 of 6,277 spells, all necessarily at the 2012 boundary). This identifies 6,277 distinct spells.

Because a tie’s sign is coded from two substantively different relational layers (positive from a ratified alliance, negative from a militarized dispute, absent when neither is present), a transition on a single edge is not simply “a sign

switching.” Table 1 makes each transition’s constituent states explicit, since a term like hostility-to-alliance realignment describes the classification, not the underlying data event.

A negative-to-positive transition — a hostility-to-alliance realignment — therefore requires both that the militarized dispute end and that a qualifying alliance exist in the following year, not merely that the dispute lapse; the reverse, an alliance-to-hostility realignment, requires both that the alliance end and that a qualifying dispute begin. Neither is a mere non-event: each is a substantive change in the relationship’s formal status, not simply the absence of renewal.

The two cases introduced above illustrate the full spell structure concretely. Britain–Germany–USSR becomes unbalanced in 1940 — three mutually hostile ties, an odd number of negatives — and contributes exactly one triad-year at risk before Germany’s June 1941 invasion of the Soviet Union is followed within weeks by the Anglo-Soviet Agreement: the Britain–USSR tie flips to positive, while Britain–Germany and Germany–USSR remain negative, and the spell ends in realignment. United States–France–USSR becomes unbalanced in 1977, with the US–France alliance and the France–USSR entente both positive while the US–USSR tie was negative — two positive ties and one negative tie, unbalanced by the Cartwright-Harary criterion. It remains a closed, unbalanced triad continuously through 1986: ten consecutive triad-years at risk of either transition. In 1987, no US–USSR dispute meeting the hostility threshold is recorded, so the triad is no longer closed; the spell ends in dissolution. Both cases are treated identically by the estimation strategy below: each contributes one triad-year row per year at risk, with that year’s structural covariates, until its observed transition.

3.2 Measuring embeddedness, actor constraint, and controls

All structural measures are computed from the full signed network of alliance and dispute ties in a given year — not from the focal triad in isolation — and are recomputed every year a spell continues, so a triad’s covariates evolve as the surrounding network changes rather than being frozen at the moment it became unbalanced.

Triadic embeddedness, the primary explanatory variable, is the mean, across a triad’s three edges, of each edge’s neighborhood embeddedness: for a given tie, the number of other states connected to both of its endpoints, equivalently the number of overlapping triangles that tie anchors. A tie

with low embeddedness — few or no common neighbors — is comparatively free-standing.

Edge betweenness centrality, a tie’s share of shortest paths between all other pairs of states in the network, captures a related but distinct property: how much a tie bridges otherwise distant parts of the network, rather than how redundantly it is locally embedded. It is tested alongside embeddedness throughout, as a separate predictor, because the two need not move together — a tie can bridge two otherwise-disconnected blocs (high betweenness) without being locally embedded in many overlapping triangles (low embeddedness), or vice versa. The results below show embeddedness, not betweenness, predicts both hazards.

Actor-level constraint is each state’s Burt (1992) network constraint, computed on the unweighted topology of the full network of that year’s alliance and dispute ties (treating a positive and a negative tie as equally strong connections, since Burt’s formula was not developed for signed weights) and averaged across the triad’s three members. Constraint is high when a state’s partners are themselves densely interconnected, leaving the state little room to play partners off against each other or exit unilaterally; it is low when partners are structurally disconnected, giving the state more independent latitude — the classic “structural holes” argument for brokerage advantage.

Capability. All models control for each triad’s mean log Composite Index of National Capability (CINC), the standard COW measure of relative material power combining six indicators — military expenditure, personnel, energy consumption, iron and steel production, population, and urban population, each as a state’s share of the contemporaneous world total — from the National Material Capabilities dataset, v7 (Singer, Bremer, and Stuckey 1972). This absorbs the possibility that the structural effects reported below simply proxy for great-power involvement.

Era. All models include fixed effects for three periods (pre-1945 / 1945–1989 / post-1989), coarse enough to avoid separation problems in a finer partition (see the online appendix) while absorbing systematic shifts in the international system’s density and character — the number of states, the prevalence of formal alliance blocs, and the base rate of militarized disputes all differ sharply across these eras.

Unbalanced load. To test H2, I construct, for every edge in every year, its *unbalanced load*: the number of other closed triads (besides the focal one) containing that edge that are themselves unbalanced that year, computed from the same triangle enumeration used to build the triad panel (Section 3.1). It is deliberately distinct from triadic

Table 1. Constituent dyad states at t and $t+1$ for a changing edge, and the transition class each combination is coded as.

State at t	State at $t+1$	Classification
Negative (MID)	Absent	Dissolution: dispute not renewed
Positive (alliance)	Absent	Dissolution: alliance not renewed
Negative (MID)	Positive (alliance)	Realignment: hostility replaced by alliance
Positive (alliance)	Negative (MID)	Realignment: alliance replaced by hostility

embeddedness, which counts every overlapping triangle regardless of whether it is balanced — a tie can be highly embedded while under little consistency pressure if most of what surrounds it is already internally consistent. Unbalanced load isolates the component of a tie's structural position that specifically reflects unresolved pressure from elsewhere in the network at that moment.

Figure 1 makes the two constructs concrete with a minimal schematic sequence, constructed for exposition rather than drawn from the data. Panels (a)–(c) build up embeddedness: a bare tie P–Q (a) becomes embedded once a third state R closes an overlapping triangle with both endpoints (b, embeddedness = 1), and further embedded as additional such triangles form (c, embeddedness = 2), independent of any tie's sign. Panels (d)–(f) then show that embeddedness and unbalanced load are not the same thing: (d) and (e) hold the P–Q and P–R ties fixed and vary only the sign of Q–R, which determines whether the same triangle is balanced (d, unbalanced load = 0) or unbalanced (e, unbalanced load = 1); (f) shares (c)'s exact topology but with both satellite ties negative, so both triangles are unbalanced and the P–Q tie's unbalanced load reaches 2. Two ties can therefore share identical embeddedness, as (c) and (f) do, yet carry very different load, depending on how many of their shared triangles are internally consistent.

Robustness controls. Three further triad-level measures serve as robustness checks, each available only for a subset of the full period. *Trade dependence* (Correlates of War Trade dataset, v4.0; Barbieri, Keshk, and Pollins 2009; Barbieri and Keshk 2016; 1870–2012) is, for each tie, the geometric mean of each state's bilateral trade with its partner as a share of that state's total trade, a proxy for how costly a relationship would be to sever economically, independent of topological embeddedness (52 of 886,828 dyad-years, 0.01%, are excluded because the recorded bilateral trade value exceeds one or both states' own recorded total trade, a mechanical impossibility concentrated in 1960s Central and Eastern European dyads, where COW's early trade-total figures are least reliable). *Geographic contiguity* (COW Direct Contiguity, v3.2; Stinnett et al. 2002) is the share of a triad's three dyads sharing a land border or narrow water separation, since proximate states are both more prone to

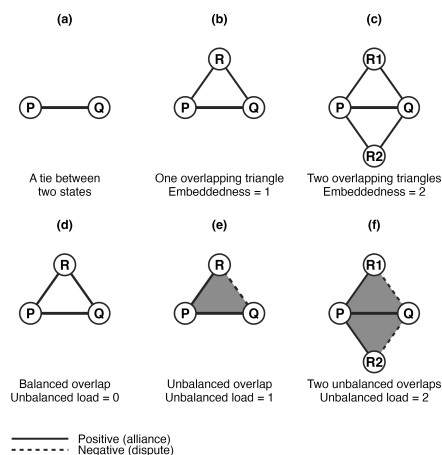


Figure 1. Triadic embeddedness and unbalanced load, illustrated schematically. (a)–(c) Embeddedness counts a tie's overlapping triangles, regardless of sign: the P–Q tie gains one overlapping triangle in (b) and two in (c). (d)–(f) Unbalanced load counts how many of those overlapping triangles are themselves unbalanced. (d) and (e) are a minimal pair holding topology fixed: the P–Q and P–R ties are unchanged, and only the sign of Q–R differs — positive in (d), where the triangle is balanced (unbalanced load = 0), negative in (e), where the same triangle is unbalanced (unbalanced load = 1). (f) shares (c)'s two-triangle topology exactly — embeddedness = 2 in both — but here both satellite ties are negative, so both triangles are unbalanced and the P–Q tie carries unbalanced load = 2. Solid lines are positive (alliance) ties; dashed lines are negative (dispute) ties.

militarized disputes and more likely to formalize an alliance. *Regime type* is the minimum Polity2 score (–10 to +10) across a triad's three members (Polity5; Marshall, Gurr, and Jaggers 2019), the standard “weakest-link” operationalization in the joint-democracy literature (e.g., Maoz and Russett 1993; Dixon 1994): a triad is only as consistently democratic as its least democratic member.

3.3 Competing-risks estimation

Competing-risks duration models are not new to international relations: Leeds and Savun (2007) use one to distinguish the different ways formal alliances end, and Quiroz Flores (2012) applies the same logic to war termination and leader survival. The contribution here is not the method but the recognition that triadic imbalance has exactly this structure — an unbalanced triad is continuously exposed to

two competing terminal events for as long as it survives, and the question of how long it persists cannot be separated from the question of which event eventually ends it.

Setup. A spell is a unit's continuous exposure to two competing, mutually exclusive terminal events, dissolution and realignment, with right-censoring at 2012 for spells still ongoing. This is a classic competing-risks design (Allison 2014): the event that ends a spell precludes the other from being observed for it, and time spent unbalanced without either event is not an outcome but time still at risk. Each spell is expanded into triad-year form, one row per calendar year at risk, carrying that year's covariates (triadic embeddedness, actor constraint, capability, and, in robustness specifications, trade dependence, contiguity, and regime type), so a spell's risk profile evolves rather than being fixed at onset. This yields 10,170 usable triad-years across 6,176 spells with at least one qualifying at-risk year: 6,097 end in an observed transition (678 dissolution, 5,419 realignment), and the remaining 79 are still-censored spells contributing only non-terminal at-risk years. Of the panel's 180 right-censored spells overall, the other 101 had only a single year at risk and contribute no rows here, since that year's transition would only have been observable in 2013, outside the panel.

Model. I take a cause-specific hazard approach: for each triad-year i at risk in year t , I model the two competing hazards separately as

$$\Pr(\text{dissolve}_{i,t+1} = 1 \mid \text{at risk}_{i,t}) = \text{logit}^{-1}(\beta_0 + \beta_1 E_{i,t} + \beta_2 C_{i,t} + \beta_3 K_{i,t} + \gamma_{\text{era}(t)})$$

and, on the same risk set, the analogous equation for $\Pr(\text{realign}_{i,t+1} = 1 \mid \text{at risk}_{i,t})$, where $E_{i,t}$ is triadic embeddedness, $C_{i,t}$ is actor constraint, and $K_{i,t}$ is capability (log CINC), each measured for triad-year i in year t . This is the discrete-time analogue of the Cox cause-specific hazard model: each cause's hazard is estimated on the full risk set, treating triad-years in which the *other* event occurs as simply not experiencing *this* event that year (coded 0) — the standard, most directly interpretable competing-risks approach for the covariate-specific rate of each transition itself, rather than the cumulative probability of that transition in the presence of the competing risk (the quantity a subdistribution-hazard, or Fine-Gray, specification targets instead). A logistic link is used rather than complementary log-log; with annual intervals and hazard probabilities well below where the two links diverge appreciably, this keeps coefficients on the same footing (log-odds) as the fixed-window robustness models. Standard errors are clustered by

triad throughout, since a given triad can contribute multiple triad-years within a spell, and sometimes multiple spells, neither independent.

Why not a fixed window. A natural alternative — asking whether a triad has resolved within, say, 5 years of becoming unbalanced — was estimated first (reported in the online appendix) and points in the same direction throughout, but the competing-risks design has two advantages that make it primary. First, it removes the need to justify an arbitrary window length, since every year of a spell is modeled on its own terms rather than collapsed into a single yes/no outcome at a fixed horizon. Second, it uses substantially more of the data: a spell lasting fifteen years contributes fifteen informative triad-year observations of the hazard it faces each year, rather than a single onset-year observation of whether it happened to resolve within an arbitrarily chosen window. The two cause-specific logits are estimated separately rather than jointly; a single multinomial discrete-time model (persist/dissolve/realign) reported in the online appendix confirms embeddedness retains the same opposing signs and statistical significance when the three outcomes are modeled jointly instead (Section A12).

3.4 Identification: what these estimates can and cannot claim

Triadic embeddedness, actor constraint, and unbalanced load are all computed from the same yearly signed network that generates the outcomes being modeled, so the hazard models throughout should be read as *structural associations conditional on a rich set of controls* — capability, era, spell duration, and, in robustness specifications, trade, contiguity, and regime type — not as causal estimates of the effect of a manipulated network property. This limitation is standard for observational network analysis and is stated here as a design feature. Two checks bound how much it matters. Re-estimating the primary models with covariates lagged a full year earlier than the (already-predetermined) year the primary specification uses leaves triadic embeddedness's effect on both hazards essentially unchanged, bounding *short-run* simultaneity — the concern that this year's covariate is itself shaped by this year's transition. A Cinelli and Hazlett (2020) omitted-variable sensitivity analysis shows an unobserved confounder would need to explain well over 10% of the residual variance in both embeddedness and the outcome, more than the paper's own strongest observed control achieves, to explain the effect away; this bound applies regardless of when such a confounder operated, but only if it resembles the observed controls in how it

relates to embeddedness and the outcome. Neither check can address the deeper, *long-run* endogeneity of network structure to decades of prior alliance and dispute politics: no observational design applied to a single historical network system can. What the checks establish is narrower but useful: the associations are not an artifact of this design's specific timing choices, nor readily explained by a confounder of ordinary strength. Full detail, including the exact coefficients and bounds, is in the online appendix (Section A4).

4. Results

4.1 Embeddedness reallocates resolution toward realignment

The trajectories are evident before any model is introduced. Figure 2 follows spells from their onset, grouped by embeddedness at the moment they became unbalanced, and shows what becomes of them over the following twenty years. Among spells beginning in the lowest embeddedness tercile, the majority end in dissolution within the first few years, with realignment accounting for a smaller share. Among spells beginning in the highest tercile, dissolution is rare and the great majority realign quickly, with almost no residual “still unbalanced” band remaining by year 10. Spells beginning at intermediate embeddedness show the largest and most persistent residual share of ongoing imbalance, exactly as the theoretical account in Section 2 predicts: embeddedness must be high enough to foreclose the low-cost exit through dissolution before it can reliably deliver the higher-effort resolution through realignment, so it is the configurations caught between the two thresholds that persist longest.

The competing-risks hazard models test whether this descriptive separation survives controlling for capability, era, and the spell's own duration, entered as a set of fixed effects for years since onset (since the cumulative-incidence pattern above suggests transitions cluster early in a spell's life). It does. As ties become more deeply embedded in the surrounding network, dissolution becomes dramatically less likely while realignment becomes more likely: triadic embeddedness reduces the hazard of dissolution ($b=-2.51$, $SE=0.16$, $p<.001$; odds ratio ≈ 0.081 per SD, roughly a 92% reduction) and increases the hazard of realignment ($b=+0.44$, $SE=0.05$, $p<.001$; odds ratio ≈ 1.55 per SD, roughly a 55% increase), reported in full in Table 2 (Figure 3). These are full-sample estimates; Section 4.5 shows the same direction holds outside major alliance blocs, but at noticeably different magnitude. Both effects are precisely estimated and change little from a specification that omits the duration fixed effects

(online appendix, Section A1); edge betweenness is a much weaker predictor in both hazards (dissolution: $b=0.03$, $p=.63$; realignment: $b=-0.03$, $p=.37$) — embeddedness specifically, not bridging position, does the work.

Section 2.2 distinguishes total embeddedness, a triad-level *exposure* measure, from unbalanced load's instantaneous *pressure* (Section 2.3); a natural question is whether the interstate network actually contains enough between-triad variation to separate them empirically. It does not. Triadic embeddedness and current unbalanced load are almost perfectly correlated at the triad-year level ($r=0.97$ across the panel's 10,170 triad-years), because the unbalanced share of a tie's overlapping triangles is remarkably stable across the embeddedness distribution (median 32%, interquartile range 31–32%) within this heavily bloc-concentrated risk set. Unsurprisingly, entering both into the realignment hazard produces unstable coefficients (VIF ≈ 15): unbalanced load is strongly positive ($b=+0.75$, $p<.001$) while embeddedness's own coefficient changes sign ($b=-0.30$, $p=.025$). These estimates cannot identify whether embeddedness's association with realignment operates through contemporaneous imbalance pressure or through a distinct exposure mechanism — conceptually the two are separate, but empirically, at this level, they are not identified apart from one another. The cleaner separation occurs within realigning triads, where total embeddedness varies little across the three constituent ties (Section 4.3 notes 74.6% of realignment events are exactly tied on it) but unbalanced load varies substantially and predicts which tie changes (Section 4.3). Full detail, including the ratio's distribution and both hazards' full coefficient tables, is in the online appendix (Section A13).

4.2 A supplementary structural result: actor constraint

Actor-level constraint (Burt 1992), estimated jointly with triadic embeddedness and reported directly in Table 2, reduces the dissolution hazard and increases the realignment hazard in the same direction as embeddedness ($b=-0.50$ and $b=+0.22$ respectively, both $p<.001$), evidence that position-level and relation-level structural constraint operate as parallel, reinforcing mechanisms once properly separated. A model that includes actor constraint but omits embeddedness recovers a coefficient of the *opposite sign* on both hazards, because the two are substantially negatively correlated ($r=-0.78$): high-degree states that broker the most overlapping relationships mechanically register *low* Burt constraint, since their many partners are not all connected to one another.

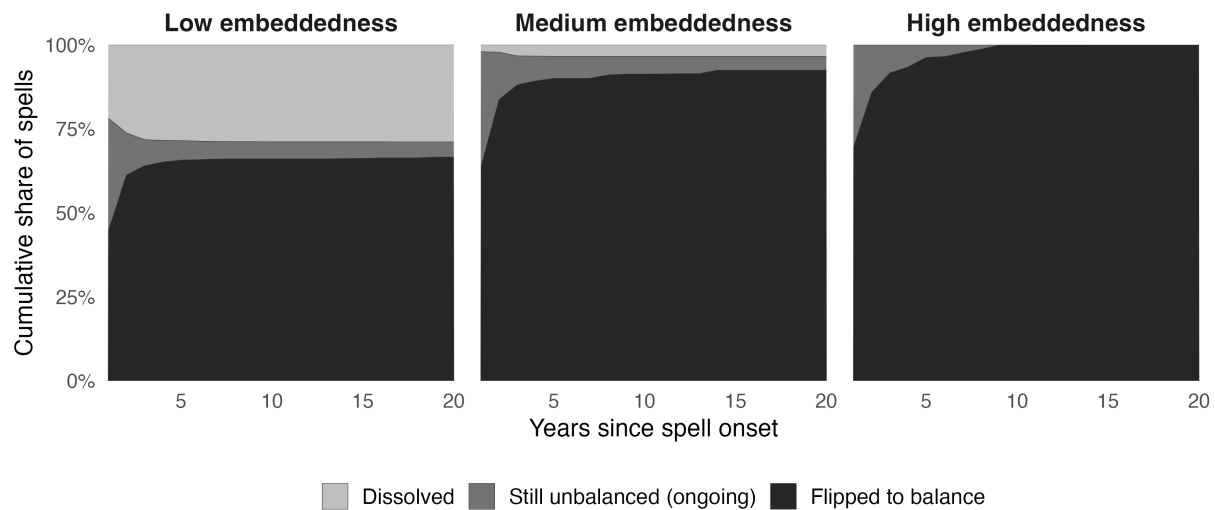


Figure 2. How unbalanced spells end over time (cumulative incidence): dissolved, still ongoing, or realigned, by embeddedness tercile at onset.

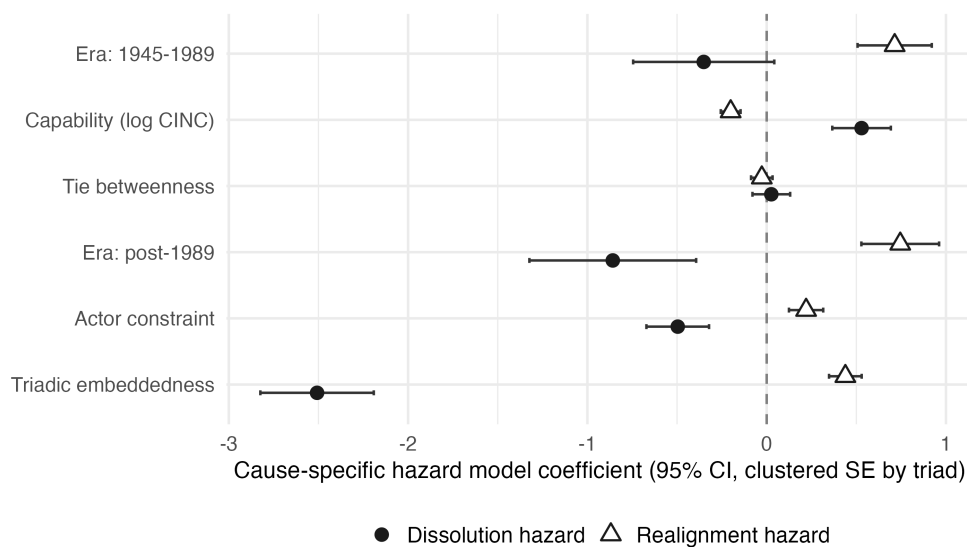


Figure 3. Cause-specific hazard model coefficients: dissolution vs. realignment.

Table 2. Discrete-time cause-specific hazard models of dissolution and realignment (logistic, clustered SEs by triad). All predictors standardized; OR = odds ratio, the substantive-scale effect of a one-SD increase. A specification without duration fixed effects, and the naive embeddedness-omitted model producing an opposite-signed actor-constraint coefficient (Section 4.2), are reported in the online appendix (Sections A1 and A6).

Predictor	Dissolution <i>b</i> (SE)	Dissolution OR	Realignment <i>b</i> (SE)	Realignment OR
Triadic embeddedness	-2.508*** (0.161)	0.08	0.439*** (0.046)	1.55
Tie betweenness	0.026 (0.054)	1.03	-0.027 (0.031)	0.97
Actor constraint	-0.496*** (0.089)	0.61	0.220*** (0.049)	1.25
Capability (log CINC)	0.529*** (0.083)	1.70	-0.201*** (0.028)	0.82
Era: 1945–1989	-0.351 (0.201)	0.70	0.714*** (0.106)	2.04
Era: post-1989	-0.858*** (0.237)	0.42	0.745*** (0.111)	2.11
Duration: year 2	-0.403** (0.148)	0.67	-0.118* (0.053)	0.89
Duration: year 3	0.827*** (0.203)	2.29	-0.938*** (0.084)	0.39
Duration: years 4–5	-1.788*** (0.406)	0.17	-1.492*** (0.086)	0.22
Duration: years 6–10	-1.750*** (0.411)	0.17	-1.872*** (0.100)	0.15
Duration: years 11+	-2.598** (0.959)	0.07	-1.737*** (0.114)	0.18
Intercept	-4.020*** (0.261)	—	-0.246* (0.104)	—
N (triad-years)	10,170		10,170	

* $p < .05$, ** $p < .01$, *** $p < .001$. Reference category for era: pre-1945; reference category for duration: spell's first year at risk.

This is a specification lesson for anyone extending the design with additional structural covariates, not a third contribution alongside H1 and H2; the naive, embeddedness-omitted comparison is reported in the online appendix (Section A6).

4.3 Within a realigning triad, the tie under the most unbalanced-triad pressure is the one that changes

H1 concerns which triads realign; H2 asks which specific tie changes when one does. Testing this requires identifying, for every realignment event, which of the triad's three ties actually changed sign — 5,414 of 5,419 realignment events (99.9%) involve exactly one, giving a clean three-way choice per event.

Unbalanced load, not total embeddedness, is the measure that does this work. Total embeddedness counts every triangle a tie is embedded in, whether or not those triangles are themselves unbalanced, and 74.6% of realignment events have exactly tied total embeddedness across all three of a triad's edges — a direct consequence of the alliance-bloc concentration documented in Section 4.5, since three states embedded in the same large multilateral pact mechanically share near-identical overlapping-triangle counts — leaving too little variation, and no significant pattern in what remains (Section 2.3 and the online appendix report this in full). Unbalanced load — the number of *other* unbalanced triads a tie is embedded in, isolating specifically the component of its position that reflects unresolved pressure — varies far more across the three edges: only 0.1% of events have it exactly tied across all three edges, leaving 2,516 events (46.5%) with all three values distinct. Among these, the tie that changes has the *highest* unbalanced load in 96.6% of events, against a 33.3% chance baseline ($p < .001$). A conditional logit confirms this with a large, precisely estimated coefficient (odds ratio ≈ 1.38 per unit of unbalanced load, $p < .001$) that is essentially unchanged when total embeddedness and betweenness are added as controls — and embeddedness's own coefficient becomes small and only marginally significant once unbalanced load is in the model.

This pooled figure needs one qualification before it is read as evidence of a general “pressure” mechanism. In the two-positive-one-negative configuration that generates the overwhelming majority of realignment events, the negative tie is definitionally the one edge shared by every unbalanced triangle in the triad, so it is structurally the likeliest candidate to carry the highest unbalanced load regardless of the specific mechanism at work; a bare indicator for “is this the negative

tie,” with no load term at all, predicts which tie changes about as well as unbalanced load does (concordance 0.977 versus 0.975). The cleaner test restricts comparisons to tie pairs that already share the same sign, where a sign indicator carries no information: among such pairs, the higher-unbalanced-load tie is still the one that changes in 75.2% of cases ($n=137$, 95% CI [67.1%, 82.2%], $p < .001$ against a 50% null). Because the same dependence that inflates the pooled figure could affect this comparison too, the 137 pairs were traced to their underlying transitions: only 71 unique dyad-year changes, and a cluster bootstrap resampling those transitions gives a 95% interval of [63.0%, 84.5%] around 75.2% — wider than the naive interval, but entirely above 50%. Unbalanced load therefore predicts tie choice beyond current sign, robust to the dyad-year dependence check above; the same-sign comparison — not the pooled 96.6% — is the more conservative estimate of that association. Full detail, including the pooled-figure dependence check and a head-to-head comparison against a net-balance-gain alternative measure, is in the online appendix (Section A12).

The results form a coherent account across three levels of analysis: triadic embeddedness predicts *whether* a triad realigns (H1); a tie's unbalanced load — its embeddedness in specifically the unresolved part of the surrounding structure — predicts, beyond what tie sign alone would predict, *which* tie realigns when it does (H2); and actor-level constraint, reported as a supplementary result in Section 4.2, operates in the same direction as triadic embeddedness once properly separated from it. Full detail on both measures is in the online appendix.

Does this pattern hold equally for alliance-to-hostility and hostility-to-alliance realignment? The tie that changes sign in a realignment event moves in one of two directions: an alliance replaced by hostility (positive-to-negative) or hostility replaced by an alliance (negative-to-positive). These are not symmetric events substantively — one is a formal commitment's failure, the other its formation — and Section 2.3 anticipated they might not be symmetric in what predicts them either. Of the 2,516 events with all three ties' unbalanced load distinct, 67 (2.7%) are alliance-to-hostility and 2,449 (97.3%) are hostility-to-alliance; the pooled 96.6% figure above is therefore overwhelmingly a statement about the latter. Splitting the test by tie type (Table 4) shows the two diverge sharply. Among hostility-to-alliance events, unbalanced load performs even better than the pooled estimate suggests: the changed tie has the highest unbalanced load in 98.9% of cases ($n=2,449$, 95% CI [98.4%, 99.3%]). Among alliance-to-hostility events, the pattern does not merely weaken — it reverses: the changed tie has the

Table 3. Conditional logit, within-triad tie choice at realignment (3 ties as alternatives, stratified by event). Fit on all 5,414 single-tie-change realignment events (the binomial test in text uses the 2,516-event subsample with all three ties' unbalanced load distinct; the conditional logit's exact partial-likelihood method handles tied cases directly).

Predictor	Unbalanced load only	With controls
Unbalanced load	0.320*** (0.008)	0.322*** (0.009)
Triadic embeddedness	—	-0.064* (0.027)
Tie betweenness	—	0.001 (0.001)
N (tie-alternatives)	16,242	16,242
N (events)	5,414	5,414
Concordance	0.975	0.977

* $p < .05$, *** $p < .001$. Concordance is the standard coxph-family statistic (the conditional logit is fit as a stratified Cox model): the share of comparable tie-pairs within an event for which the model ranks the higher-unbalanced-load tie more likely to change.

highest unbalanced load in only 11.9% of cases ($n=67$, 95% CI [5.3%, 22.2%]), significantly *below* the 33.3% chance baseline. A two-sample test of proportions confirms this gap is not sampling noise ($\chi^2=1467.5$, $df=1$, $p<.001$; 95% CI for the difference in hit rates, alliance-to-hostility minus hostility-to-alliance: [-95.5, -78.4] percentage points).

The reversal is not an artifact of alliance-to-hostility events starting from a different triad configuration: a direction-of-change cross-tab (Table 5) shows that essentially all alliance-to-hostility events (67 of 67) begin, like the hostility-to-alliance events, from the same two-positive-one-negative starting configuration; the tiny residual zero-positive-three-negative configuration (5 events total) contributes only to the hostility-to-alliance side and is too small to explain the reversal. Whatever distinguishes the two tie types, it is not which triad shape they start from.

A conditional logit with an unbalanced-load $\times$ tie-type interaction term, fit on the full 5,414-event sample (16,242 tie-alternatives, matching Table 3's primary specification), estimates the interaction itself as small and not statistically distinguishable from zero ($b=-0.023$, $SE=0.044$, $p=.60$; likelihood-ratio test: $\chi^2=0.28$, $df=1$, $p=.60$), even though the tie-type main effect is large and significant on its own ($b=-2.44$, $SE=0.14$, $p<.001$). These results are best read as a striking descriptive boundary condition (Table 4) that the data cannot formally establish as a moderation effect: unbalanced load predicts which tie changes almost deterministically for the dominant transition type, hostility-to-alliance, while the much smaller alliance-to-hostility subsample shows a markedly different, reversed pattern. Both transitions are substantive relational reclassifications, not a mere non-event on either side, so the asymmetry needs a different account than "commitment failure versus non-event." Why it occurs cannot be established with the present data; one possibility is that transitions into hostility face additional bilateral and domestic constraints that transitions toward alignment do not (Leeds and Savun 2007; Fearon 1994), which would require dyad-specific

information beyond the structural measures used here to test. The pooled 96.6% figure and Table 3's conditional logit are accordingly best read as a strong, precisely estimated finding about hostility-to-alliance realignment specifically, carried by its much larger share of the realignment events observed, not a general law confirmed to hold across both directions of change.

4.4 The modal dissolution event is a dispute lapsing, not an alliance expiring or a state disappearing

Treating "dissolution" as a single category risks pooling substantively different processes — an alliance reaching the end of its term, a militarized dispute simply not recurring, or one of the two states involved exiting the interstate system entirely (conquest, unification, dissolution of the state itself). Disaggregating the 678 dissolution events resolves part of the concern: 60.0% (407) involve exactly one of the triad's three ties disappearing, and among these, 80.1% are a militarized dispute not recurring and only 19.9% are an alliance not renewed; none coincide with either endpoint state exiting the system. The remaining 40% of dissolution events (271) involve two or three ties disappearing simultaneously; only 6.6% of these coincide with a state exiting the system, but which tie type is responsible is not separately broken down for this multi-tie subset, so the 80.1% dispute-lapse figure is established for the single-tie majority specifically, not for dissolution as a whole. Taken together, the modal dissolution event is nonetheless specifically a militarized dispute not being renewed — closer to what the rivalry- and alliance-termination literatures already study under different names (Leeds and Savun 2007 on alliance abrogation specifically) than to alliance expiration or state death.

This is not the same claim as "rivalry termination" in the sense that literature uses the term. A rivalry is a longer-running relational construct that can encompass many separate MIDs; a single dispute not recurring is

Table 4. Unbalanced-load hit rate (changed tie has the maximum value among the triad's three ties), by changed-tie type. Restricted to the 2,516 events with all three ties' unbalanced load distinct; null hit rate under chance is 33.3%.

Group	N	Hit rate	95% CI	p (vs. 1/3 baseline)
All events	2,516	96.6%	[95.8%, 97.3%]	<.001
Alliance-to-hostility (pos → neg)	67	11.9%	[5.3%, 22.2%]	<.001
Hostility-to-alliance (neg → pos)	2,449	98.9%	[98.4%, 99.3%]	<.001

Table 5. Direction-of-change cross-tab: triad configuration at the start of the realignment year, by changed-tie type.

Triad configuration (t)	Changed tie type	N	Hit rate
Two positive, one negative	Alliance-to-hostility (pos → neg)	67	11.9%
Two positive, one negative	Hostility-to-alliance (neg → pos)	2,444	99.0%
Zero positive, three negative	Hostility-to-alliance (neg → pos)	5	60.0%

at most one data point toward, not equivalent to, the end of an underlying rivalry, and an old methodological literature has debated at length whether a gap in MID data reliably indicates genuine termination rather than a lull within a continuing rivalry (Bennett 1997). The negative-tie coding used here is deliberately dispute-level, since that is what the triad-year design requires; whether a given dispute's lapse here coincides with the termination of the broader rivalry it belongs to is a question this design does not adjudicate. Section 5 treats the connection to rivalry-termination research accordingly, as a neighboring implication rather than as a claim that this paper directly measures when rivalries end.

4.5 Robustness

Trade dependence, available for the 1870–2012 subsample ($n=9,811$ triad-years), is a significant predictor in both hazards alongside topological embeddedness, which retains its full strength: dissolution hazard, trade $b=-0.23$ ($p=.006$), embeddedness $b=-2.46$ ($p<.001$); realignment hazard, trade $b=+0.17$ ($p<.001$), embeddedness $b=+0.37$ ($p<.001$). Economic and topological embeddedness point the same way and neither absorbs the other's effect.

Adding geographic contiguity and regime type (restricted subsample with Polity5 coverage, $n=8,076$ triad-years, 1850–2011) leaves the primary results intact: embeddedness remains large and highly significant in both hazards (dissolution $b=-2.60$, $p<.001$; realignment $b=+0.29$, $p<.001$), as does actor constraint (dissolution $b=-0.36$, $p=.003$; realignment $b=+0.14$, $p=.008$). Both added controls are themselves informative, though contiguity's role differs by hazard and regime type's dissolution-hazard estimate is weaker than without duration control. Contiguity is negative and significant on the realignment hazard ($b=-0.08$, $p=.003$), plausibly because geographic proximity raises the salience of a relationship enough to make re-signing harder, but is not significant for dissolution once duration is modeled ($b=-0.08$,

$p=.25$). Regime type (minimum Polity2 across the triad) is positive and significant on realignment ($b=0.14$, $p<.001$); on dissolution it is negative but no longer even marginally significant once spell duration is controlled ($b=-0.12$, $p=.15$), a genuine attenuation worth flagging rather than glossing over. Neither control changes the sign or substantially changes the magnitude of the embeddedness or actor-constraint effects; all coefficients here, as in Table 2, are estimated with spell-duration fixed effects included (online appendix, Section A1).

A fixed 5-year-window logistic decomposition, matching the outcome coding used before adopting the competing-risks design, reproduces the same substantive pattern and is reported in the online appendix for readers who prefer a more familiar fixed-window benchmark.

One further check addresses the international alliance system's concentration into a small number of major 20th- and 21st-century blocs (NATO, the Warsaw Pact, the Rio Pact, and others): 87.6% of the triad-years analyzed here fall within one of these blocs at the time. This concentration is also a dependence problem, not only a generalizability one: a single tie changing sign or disappearing inside a large bloc can end many overlapping triadic spells at once, so triad-clustered standard errors alone understate how much shared information the panel's 6,277 nominal spells actually carry (Section 5 discusses this limitation directly). The two-way-clustered-by-triad-and-bloc specification below is this check's answer to that dependence concern, not only to bloc concentration as a substantive feature of the alliance system. The embeddedness effect survives this concentration, correctly signed and significant whether bloc-touching spells are excluded, bloc membership is added as a control, or standard errors use two-way clustering by triad and bloc, though the two-way-clustered standard errors roughly double, and part of the effect's magnitude reflects the exceptional durability of formal defense pacts specifically. A complementary check reweights embeddedness so a shared

membership in a large, diffuse alliance counts for less than one in a small, exclusive pact; this alternative measure correlates negatively with the primary one ($r=-0.42$) yet still points the same direction, complementary rather than competing when entered together. Full detail in the online appendix (Sections A2-A3).

5. Discussion

Embeddedness supplies the explanation this paper set out to test: weakly embedded inconsistencies mostly disappear, strongly embedded ones mostly get realigned, and it is only the intermediate range where genuine multi-year persistence concentrates. Particular dynamic formulations of balance theory treat a tie as a fixed feature of the relationship space that can only be flipped (Antal, Krapivsky, and Redner 2005; Marvel et al. 2011) — a modeling choice within those formulations, not an inherent commitment of balance theory as such (Section 2.1); allowing for dissolution, an exit option those particular models omit entirely, is what distinguishes which trajectory a given episode of imbalance takes. A substantial share of this evidence comes from ties embedded inside major multilateral alliance blocs (NATO, the Warsaw Pact, the Rio Pact); Section 4.5 shows the direction holds outside those blocs too, though its magnitude is inseparable from the institutionalized durability such blocs confer. The two mechanisms decompose unevenly: outside blocs, embeddedness's effect on dissolution is smaller ($b=-0.78$) but its effect on realignment is if anything *larger* ($b=+1.10$) than the full-sample estimate, while inside blocs the pattern reverses (effective $b=-2.03$ on dissolution, $b=+0.33$ on realignment; online appendix, Section A2). Formal multilateral institutionalization mainly amplifies how durable a tie is once embedded, not whether embeddedness drives realignment specifically.

Three implications follow. First, for balance theory itself: balance is not simply produced by sign-switching. Particular dynamic formulations model every unbalanced triad as eventually resolving through a flipped tie, but a large share of the triads studied here resolve instead by a tie disappearing, and which happens is systematically predicted by structure — down to which specific tie moves within a realigning triad (Section 4.3), extending Rawlings and Friedkin's (2017) balance-violation exposure mechanism to a within-event choice among a triad's three relationships, where total embeddedness fails and unbalanced load succeeds, concentrated in one direction of change. The actor-constraint result (Section 4.2) adds a related point: a pair of correlated measures needs to be estimated jointly, since dropping

either recovers a misleadingly reversed coefficient on the other. Second, for what "persistence" means: a configuration that looks anomalously durable in a model tracking only realignment is not necessarily evidence that balance pressure has failed. It can instead occupy the structural region between a viable exit and a viable adaptation — embedded enough that dissolution is no longer low-cost, but not so embedded that realignment is yet reliably available; persistence marks where a configuration sits between two resolution regimes, not a residual category left over once balance-restoring pressure is assumed to have failed.

Third, for international relations specifically: this connects the triadic, network-structural logic of balance theory to two distinct bodies of research that have each captured part of the picture without the other half. IR network research (Warren 2010; Cranmer, Desmarais, and Menninga 2012; Corbetta and Grant 2012; Kinne 2024) has established that alliance formation and dispute intervention are extra-dyadically interdependent, but not how that structure allocates an existing episode of imbalance between exit and realignment. Dyadic exit research — rivalry termination (Bennett 1996; Diehl and Goertz 2000; Colaresi, Rasler, and Thompson 2007; Prins and Daxecker 2008; Sakuwa and Thompson 2019; Thompson 2022) and alliance termination (Leeds and Savun 2007) — has established that both kinds of relationship do end, but without asking what triadic structure the relationship sits in. This paper's contribution sits at the intersection: embeddedness is a structural predictor of which of the two dyadic exits a given episode of inconsistency takes, and of which specific tie carries the realignment when that is the exit taken. These are related but distinct claims: H1 concerns variation between triads in the route out of imbalance, whereas H2 concerns variation among ties within triads that realign. It also complements Kinne and Maoz's (2023) finding that imbalance in a state's *domestic* political network destabilizes its foreign relations, suggesting an analogous mechanism through the state's own interstate ties. For rivalry termination specifically, dense triadic embeddedness is a plausible structural precondition for the kind of dramatic bilateral realignment that can end a rivalry outright (the introduction's Britain-USSR case); this is offered as a connection to that literature, not as a claim tested here, since the present analysis operates at the level of the individual militarized dispute rather than the longer-running rivalry construct a change of that kind would need to terminate (Section 4.4; Bennett 1997). For alliance termination, abrogation likely grows structurally costly inside a dense multilateral bloc because exit disturbs

every relationship built on the same foundation, a network-structural amendment to Leeds and Savun's (2007) dyadic account. A tie's embeddedness should predict not just *whether* it ends, but *how* — a natural bridge for future work adding dyad-specific covariates (issue salience, prior war, alliance terms) to the structural measures used here, and, given how sharply alliance-to-hostility and hostility-to-alliance realignment diverge in Section 4.3, a bridge that likely needs to treat “realignment” as two distinct processes — making and breaking a formal commitment — rather than one.

Several limitations bound these conclusions. Unbalanced load is measured on the same observed network used to define the triads and spells themselves, rather than through a design holding network structure exogenous to the dynamics explained. At the edge level used in H2, it is also, by construction, correlated with total embeddedness ($r=0.33$) and with actor-level constraint — a different quantity from the much higher triad-year-level correlation ($r=0.97$) discussed in Section 4.1, since the two are computed on different units and different samples (individual edges within realignment events versus triad-year means across the full risk set). The conditional logit shows its effect survives controlling for both, but an independent test on a different signed network would strengthen confidence the mechanism is specifically about unresolved pressure rather than general local density.

H1's dependence structure raises an analogous concern to the one just addressed directly for H2 in Section 4.3: a single tie changing sign or disappearing can end many overlapping triadic spells simultaneously, and clustering standard errors by triad does not by itself capture dependence induced by shared dyads, shared states, or common alliance-bloc membership. Section 4.5's two-way-clustered-by-triad-and-bloc specification is the H1 analogue of that check: embeddedness's effect on both hazards remains correctly signed and significant once dependence at the bloc level, not only the triad level, is modeled explicitly, though the resulting standard errors roughly double. As with H2, this bounds rather than eliminates the concern that the panel's 6,277 nominal spells overstate the amount of independent information behind the reported estimates.

The triad population is necessarily built from realized alliance and dispute ties, so small or peripheral states with no signed relationships in a given period are invisible to the analysis. Every measure here is also built on a one-mode projection that treats a multilateral alliance like a bilateral treaty; Appendix A2 shows the results are robust to, but not entirely independent of, the resulting concentration

in a handful of major blocs. A two-mode representation, with organizations as a second node type, would be more faithful, closer to Heider's (1946) original formulation, and a proxy for it (Appendix A3) points the same direction; Fritz et al.'s (2025) dynamic signed exponential-random-graph model offers one plausible framework for the full redesign.

The H2 heterogeneity between alliance-to-hostility and hostility-to-alliance realignment rests on a small alliance-to-hostility subsample ($n=67$); the descriptive gap is large, but the formal interaction test cannot confirm it as moderation on its own, and a larger sample — plausibly requiring data beyond the interstate system — would be needed to establish it as a general property rather than a suggestive pattern here.

Both the embeddedness and actor-constraint results, while robust across every specification tested here (fixed-window and competing-risks alike, with and without duration, trade, contiguity, and regime-type controls), were developed on the same historical dataset; independent replication on signed networks outside the interstate system would strengthen confidence that “exit versus realignment” is a general property of relational systems under structural constraint, not one specific to interstate data.

None of this changes the paper's central empirical claim: unbalanced triads in the interstate system exit through dissolution or realignment, the route taken varies systematically with triadic embeddedness, and getting this right requires recognizing exit as a theoretically meaningful outcome rather than folding it into an undifferentiated “did not resolve” category. Some hostile ties end because nothing held them in place — overwhelmingly, the evidence here suggests, because a militarized dispute was simply not renewed, not because a state disappeared or an alliance's term expired. Others end in realignment because too much held them in place for exit to be an option. The apparent persistence in between is not a third kind of ending — it is simply time still at risk of one of the other two. Persistence can therefore emerge without any mechanism that positively sustains imbalance: it arises because the two processes that terminate imbalance dominate in different parts of the structural landscape, a claim about how competing exits interact that does not depend on the interstate system specifically and should travel to other relational systems under structural constraint.

Data Availability Statement

Replication code and data for this article are available at <https://anonymous.4open.science/r/two-paths-out-of-imbalance/>, an anonymized

mirror (for double-blind review) of a public repository. The package is self-contained: it includes the full analysis pipeline (25 R scripts, run in numeric order), every intermediate data panel and model output the paper reports, and the raw source data as downloaded, unmodified. All raw data are publicly available from the Correlates of War Project (Formal Alliances v4.1, Dyadic MIDs v4.03, Direct Contiguity v3.2, National Material Capabilities v7, State System Membership 2024, and Trade v4.0; correlatesofwar.org) and the Polity5 project (systemicpeace.org); exact dataset versions and download pages are listed in the repository's README.

References

- Allison, Paul D. 2014. *Event History and Survival Analysis*, 2nd ed. Sage.
- Antal, T., P. L. Krapivsky, and S. Redner. 2005. "Dynamics of Social Balance on Networks." *Physical Review E* 72(3): 036121.
- Barbieri, Katherine, and Omar Keshk. 2016. *Correlates of War Project Trade Data Set Codebook, Version 4.0*.
- Barbieri, Katherine, Omar M. G. Keshk, and Brian Pollins. 2009. "Trading Data: Evaluating Our Assumptions and Coding Rules." *Conflict Management and Peace Science* 26(5): 471–491.
- Bennett, D. Scott. 1996. "Security, Bargaining, and the End of Interstate Rivalry." *International Studies Quarterly* 40(2): 157–183.
- Bennett, D. Scott. 1997. "Measuring Rivalry Termination, 1816–1992." *Journal of Conflict Resolution* 41(2): 227–254.
- Burt, Ronald S. 1992. *Structural Holes: The Social Structure of Competition*. Harvard University Press.
- Caimo, Alberto, and Isabella Gollini. 2026. "Separable Models for Dynamic Signed Networks." *Journal of the Royal Statistical Society, Series A*, advance access. DOI: 10.1093/jrsssa/qnag053.
- Cartwright, Dorwin, and Frank Harary. 1956. "Structural Balance: A Generalization of Heider's Theory." *Psychological Review* 63(5): 277–293.
- Cinelli, Carlos, and Chad Hazlett. 2020. "Making Sense of Sensitivity: Extending Omitted Variable Bias." *Journal of the Royal Statistical Society, Series B (Statistical Methodology)* 82(1): 39–67.
- Colaresi, Michael P., Karen Rasler, and William R. Thompson. 2007. *Strategic Rivalries in World Politics: Position, Space and Conflict Escalation*. Cambridge University Press.
- Coleman, James S. 1988. "Social Capital in the Creation of Human Capital." *American Journal of Sociology* 94: S95–S120.
- Corbetta, Renato, and Keith A. Grant. 2012. "Intervention in Conflicts from a Network Perspective." *Conflict Management and Peace Science* 29(3): 314–340.
- Cranmer, Skyler J., Bruce A. Desmarais, and Justin H. Kirkland. 2012. "Toward a Network Theory of Alliance Formation." *International Interactions* 38(3): 295–324.
- Cranmer, Skyler J., Bruce A. Desmarais, and Elizabeth J. Menninga. 2012. "Complex Dependencies in the Alliance Network." *Conflict Management and Peace Science* 29(3): 279–313.
- Davis, James A. 1967. "Clustering and Structural Balance in Graphs." *Human Relations* 20(2): 181–187.
- Diaz-Diaz, Fernando, Paolo Bartesaghi, and Ernesto Estrada. 2024. "Mathematical Modeling of Local Balance in Signed Networks and Its Applications to Global International Analysis." *Journal of Applied Mathematics and Computing* 70: 6195–6218.
- Diehl, Paul F., and Gary Goertz. 2000. *War and Peace in International Rivalry*. University of Michigan Press.
- Dixon, William J. 1994. "Democracy and the Peaceful Settlement of International Conflict." *American Political Science Review* 88(1): 14–32.
- Doreian, Patrick, and David Krackhardt. 2001. "Pre-Transitive Balance Mechanisms for Signed Networks." *Journal of Mathematical Sociology* 25(1): 43–67.
- Doreian, Patrick, and Andrej Mrvar. 2015. "Structural Balance and Signed International Relations." *Journal of Social Structure* 16: 1–49.
- Estrada, Ernesto. 2019. "Rethinking Structural Balance in Signed Social Networks." *Discrete Applied Mathematics* 268: 70–90.
- Fearon, James D. 1994. "Domestic Political Audiences and the Escalation of International Disputes." *American Political Science Review* 88(3): 577–592.
- Fritz, Cornelius, Marius Mehrl, Paul W. Thurner, and Göran Kauermann. 2025. "Exponential Random Graph Models for Dynamic Signed Networks: An Application to International Relations." *Political Analysis* 33: 211–230.
- Gallo, Anna, Diego Garlaschelli, Renaud Lambiotte, Fabio Saracco, and Tiziano Squartini. 2024. "Testing Structural Balance Theories in Heterogeneous Signed Networks." *Communications Physics* 7: 154.
- Gibler, Douglas M. 2009. *International Military Alliances, 1648–2008*. CQ Press.
- Granovetter, Mark. 1985. "Economic Action and Social Structure: The Problem of Embeddedness." *American Journal of Sociology* 91(3): 481–510.
- Harrigan, Nicholas M., Giuseppe Labianca, and Filip Agneessens. 2020. "Negative Ties and Signed Graphs Research: Stimulating Research on Dissociative Forces in Social Networks." *Social*

- Networks* 60: 1–10.
- Heider, Fritz. 1946. “Attitudes and Cognitive Organization.” *Journal of Psychology* 21(1): 107–112.
- Hummon, Norman P., and Patrick Doreian. 2003. “Some Dynamics of Social Balance Processes: Bringing Heider Back Into Balance Theory.” *Social Networks* 25(1): 17–49.
- Kinne, Brandon J. 2024. “Network Context and the Effectiveness of International Agreements.” *International Studies Quarterly* 68(3): sqae099.
- Kinne, Brandon J., and Zeev Maoz. 2023. “Local Politics, Global Consequences: How Structural Imbalance in Domestic Political Networks Affects International Relations.” *Journal of Politics* 85(1): 107–124.
- Krackhardt, David. 1998. “Simmelian Ties: Super Strong and Sticky.” In *Power and Influence in Organizations*, ed. Roderick M. Kramer and Margaret A. Neale, 21–38. Thousand Oaks, CA: Sage.
- Larson, Jennifer M. 2021. “Networks of Conflict and Cooperation.” *Annual Review of Political Science* 24: 89–107.
- Leeds, Brett Ashley, and Burcu Savun. 2007. “Terminating Alliances: Why Do States Abrogate Agreements?” *Journal of Politics* 69(4): 1118–1132.
- Lerner, Jürgen. 2016. “Structural Balance in Signed Networks: Separating the Probability to Interact from the Tendency to Fight.” *Social Networks* 45: 66–77.
- Leskovec, Jure, Daniel Huttenlocher, and Jon Kleinberg. 2010. “Signed Networks in Social Media.” *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 1361–1370.
- Maoz, Zeev, Lesley G. Terris, Ranan D. Kuperman, and Ilan Talmud. 2007. “What Is the Enemy of My Enemy? Causes and Consequences of Imbalanced International Relations, 1816–2001.” *Journal of Politics* 69(1): 100–115.
- Maoz, Zeev, and Bruce Russett. 1993. “Normative and Structural Causes of Democratic Peace, 1946–1986.” *American Political Science Review* 87(3): 624–638.
- Maoz, Zeev, Paul L. Johnson, Jasper Kaplan, Fiona Ogunkoya, and Aaron P. Shreve. 2019. “The Dyadic Militarized Interstate Disputes (MIDs) Dataset Version 3.0: Logic, Characteristics, and Comparisons to Alternative Datasets.” *Journal of Conflict Resolution* 63(3): 811–835.
- Marshall, Monty G., Ted Robert Gurr, and Keith Jagers. 2019. *Polity5: Political Regime Characteristics and Transitions, 1800–2018*. Center for Systemic Peace.
- Marvel, Seth A., Jon Kleinberg, Robert D. Kleinberg, and Steven H. Strogatz. 2011. “Continuous-Time Model of Structural Balance.” *Proceedings of the National Academy of Sciences* 108(5): 1771–1776.
- Nieman, Mark David, and Douglas M. Gibler. 2023. “Peaceful Neighborhoods and Democratic Differences.” *Journal of Politics* 85(1): 92–106.
- Oishi, Koji, and Kentaro Sakuwa. 2025. “Structural Balance of Alliance and Rivalry Networks in International Relations.” *Artificial Life and Robotics* 30: 303–309.
- Polidoro, Francisco, Jr., Gautam Ahuja, and Will Mitchell. 2011. “When the Social Structure Overshadows Competitive Incentives: The Effects of Network Embeddedness on Joint Venture Dissolution.” *Academy of Management Journal* 54(1): 203–223.
- Prins, Brandon C., and Ursula E. Daxecker. 2008. “Committed to Peace? Liberal Institutions and the Termination of Rivalry.” *British Journal of Political Science* 38(1): 17–34.
- Quiroz Flores, Alejandro. 2012. “A Competing Risks Model of War Termination and Leader Change.” *International Studies Quarterly* 56(4): 809–819.
- Rawlings, Craig M., and Noah E. Friedkin. 2017. “The Structural Balance Theory of Sentiment Networks: Elaboration and Test.” *American Journal of Sociology* 123(2): 510–548.
- Sakuwa, Kentaro, and William R. Thompson. 2019. “On the Origins, Persistence and Termination of Spatial and Positional Rivalries in World Politics: Elaborating a Two-Issue Theory of Conflict Escalation.” *International Area Studies Review* 22(3): 203–225.
- Sarkees, Meredith Reid, and Frank Wayman. 2010. *Resort to War: A Data Guide to Inter-State, Extra-State, Intra-State, and Non-State Wars, 1816–2007*. CQ Press.
- Saunders, Richard J. 2023. “A Certain Gamble: Institutional Change, Leader Turnover, and Their Effect on Rivalry Termination.” *Conflict Management and Peace Science* 40(3): 281–303.
- Singer, J. David, Stuart Bremer, and John Stuckey. 1972. “Capability Distribution, Uncertainty, and Major Power War, 1820–1965.” In *Peace, War, and Numbers*, ed. Bruce Russett, 19–48. Sage.
- Singer, J. David, and Melvin Small. 1972. *The Wages of War, 1816–1965: A Statistical Handbook*. New York: John Wiley and Sons.
- Stinnett, Douglas M., Jaroslav Tir, Philip Schafer, Paul F. Diehl, and Charles Gochman. 2002. “The Correlates of War Project Direct Contiguity Data, Version 3.” *Conflict Management and Peace Science* 19(2): 58–66.
- Thompson, William R. 2022. “Constructing a General Model Accounting for Interstate Rivalry Termination.” In William R. Thompson, Kentaro Sakuwa, and Prashant Hosur Suhas, *Analyzing Strategic Rivalries in World Politics: Types of Rivalry, Regional Variation, and Escalation/De-escalation*,

ch. 13. Palgrave Macmillan.

- Uzzi, Brian. 1997. "Social Structure and Competition in Interfirm Networks: The Paradox of Embeddedness." *Administrative Science Quarterly* 42(1): 35–67.
- van de Rijt, Arnout. 2011. "The Micro-Macro Link for the Theory of Structural Balance." *Journal of Mathematical Sociology* 35(1–3): 94–113.
- Warren, T. Camber. 2010. "The Geometry of Security: Modeling Interstate Alliances as Evolving Networks." *Journal of Peace Research* 47(6): 697–709.